
SENTINEL-GNSS

Proactive Multi-Horizon Prediction of GNSS Signal
Degradation for Autonomous Vehicle Navigation

Team Pilot Project — General Report

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Abstract

Reliable absolute positioning is a safety-critical requirement for autonomous vehicles. Global Navigation Satellite System (GNSS) receivers provide this positioning but are vulnerable to silent degradation in urban environments due to Non-Line-of-Sight (NLOS) multipath and signal blockage from buildings. Existing monitoring approaches—RTKLIB quality codes, RAIM, and recent machine-learning classifiers—are *reactive*: they report degradation only after it has affected positioning performance, giving the vehicle zero preparation time.

This report describes **SENTINEL-GNSS**, a Transformer–LSTM neural network that *predicts* GNSS signal degradation 5, 15, and 30 seconds before it occurs. The model processes a 60-second sliding window of 37 receiver-derived features and outputs a three-class forecast (**CLEAN** / **WARNING** / **DEGRADED**) for each prediction horizon in a single forward pass.

Trained on 149,662 labelled epochs from Beihang University field collections and the UrbanNav Hong Kong dataset, the model achieves a Macro-F1 of **0.821** at the +5s horizon with a DEGRADED-class recall of **0.847**. Critically, when evaluated on the fully held-out city of Tokyo, SENTINEL retains a DEGRADED-class F1 of **0.75** where the strongest classical baseline (RandomForest) collapses to **0.15**, demonstrating transferable urban degradation representations.

The prediction is integrated into a 9-state Extended Kalman Filter (EKF) via adaptive measurement noise covariance inflation, producing a 48.8 % reduction in positioning RMSE during degraded epochs on the Trimble Shinjuku dataset. A Huber-robust EKF and a Bootstrap Particle Filter with Student-*t* GPS likelihood are also evaluated, with the Particle Filter achieving a 40.1 % degraded-segment gain on the Odaiba open-harbour scenario. A live, multilingual dashboard (5 languages) visualises all fusion tracks in real time.

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1 Introduction

Autonomous vehicles (AVs) depend on accurate, continuous positioning to execute safety-critical decisions at road speed. A vehicle travelling at 60 km/h covers 17 metres per second; a positioning gap of even 5 seconds can place a vehicle in the wrong lane, miss a junction, or trigger emergency braking at the wrong location. Among the positioning sensors available to AVs—cameras, LiDAR, radar, and Inertial Measurement Units (IMU)—GNSS remains the sole source of drift-free absolute position [1].

The fundamental problem is that GNSS *fails silently* in dense urban environments. NLOS multipath caused by building reflections introduces position errors of 10–200 m into the receiver solution, yet the receiver continues to report a “valid” fix with no indication of the error magnitude [2]. Existing quality monitors—RTKLIB solution quality codes, Receiver Autonomous Integrity Monitoring (RAIM) [3], and recent supervised classifiers [4, 5]—are reactive: they identify degradation at the moment it begins. At 60 km/h this gives the vehicle controller zero preparation time.

1.1 Contributions of This Work

This paper makes the following contributions:

1. **Problem reformulation.** We reframe GNSS quality monitoring as a *predictive* rather than reactive problem, and demonstrate that 5–30 s lead times are achievable with a learned sequence model.
2. **SENTINEL-GNSS architecture.** A unified Transformer–BiLSTM network with three output heads (+5 s, +15 s, +30 s) achieves Macro-F1 = 0.821 at the +5 s horizon using 1.46 M parameters and runs at 0.039 ms per sample.
3. **Cross-city generalisation.** Training on Beihang + Hong Kong and evaluating on fully held-out Tokyo, the network maintains DEGRADED-class F1 = 0.75 while the strongest classical baseline (RandomForest) drops to 0.15.
4. **Prediction-informed sensor fusion.** Integration of SENTINEL predictions into an adaptive EKF yields a 48.8 % degraded-RMSE improvement on a clean dual-frequency receiver (Trimble Shinjuku), 25.2 % on a single-frequency u-blox receiver (Huber EKF), and 40.1 % on an open-harbour scenario (Student-*t* PF).
5. **Open benchmark.** A multi-city, multi-receiver labelled dataset of 149,662 epochs across 4 cities and 9+ receiver types is released with the paper.

2 Related Work

2.1 GNSS Quality Monitoring

Traditional GNSS integrity monitoring is based on RAIM [3], which uses pseudorange consistency checks to detect and isolate faulty satellites. RAIM is well-understood and widely deployed but operates purely on the current epoch; it provides no forward-looking indication of approaching degradation. RTKLIB [6] assigns solution quality codes ($Q=1$ fix, $Q=5$ float) which are commonly used as degradation indicators in research, but these again reflect current rather than future quality.

2.2 Machine Learning for GNSS

Chen *et al.* [7] demonstrated that recurrent networks could classify GNSS signal quality using C/N_0 time series, achieving accuracy above 90 % on simulated data. Liu *et al.* [4] applied gradient-boosted trees to multi-feature GNSS classification in Hong Kong urban environments, reporting strong in-domain results. These works are *contemporaneous classifiers*: they label the current epoch as clean or degraded. SENTINEL differs in that it predicts the class of a *future* epoch, turning quality monitoring into a forecasting problem.

2.3 Adaptive Kalman Filtering

The Mehra (1970) innovation-based adaptive R estimator [8] is a classical approach to unknown measurement noise covariance. However, in urban NLOS environments it fails catastrophically: GPS and the filter co-drift, producing artificially small innovations that cause R to shrink, increasing Kalman gain and accelerating drift (we measured $RMSE > 238,000$ m in testing). Prediction-based R inflation, as proposed here, sidesteps this identification problem by conditioning R adaptation on a learned classifier output rather than the innovation sequence.

3 Dataset and Feature Engineering

3.1 Data Sources

The training dataset comprises three sources:

1. **Beihang field collections.** Five controlled scenarios (A–E) collected on campus using a Septentrio Mosaic-X5C receiver: (A) instantaneous signal blockage, (B) urban canyon traverse, (C) partial tree canopy obstruction, (D) open-sky baseline, (E) gradual building approach.
2. **UrbanNav Hong Kong.** Public dataset from Hong Kong Polytechnic University [9] covering Medium, Deep, Harsh, and Tunnel urban environments with 9+ receiver types and RTK ground truth at centimetre accuracy.

3. **UrbanNav Tokyo (held-out)**. Shinjuku (Trimble RTKLIB SPP + u-blox F9P) and Odaiba (u-blox F9P) drives, excluded from all training and validation. Used exclusively for cross-city evaluation and EKF experiments.

The combined labelled dataset contains **149,662 epochs** across 4 cities and 9+ receiver types. The three-class label schema (CLEAN / WARNING / DEGRADED) was applied uniformly: CLEAN when positioning error < 2 m and mean $C/N_0 > 35$ dB-Hz; DEGRADED when error > 5 m, $C/N_0 < 30$ dB-Hz, or satellite count < 4 ; WARNING otherwise.

3.2 Feature Engineering

Raw RINEX and NMEA observables are processed through RTKLIB [6] to produce **37 features** per epoch in seven groups:

Position quality (4) Horizontal/vertical position error, PDOP, HDOP.

Signal strength (8) Mean C/N_0 , per-satellite C/N_0 statistics, variability.

Satellite geometry (6) Satellite count, elevation angle statistics, GDOP.

Pseudorange (7) Residuals, delta-pseudorange, multipath indicators.

Temporal patterns (5) Rolling mean/std of C/N_0 and error over 10-s window.

Atmospheric (4) Tropospheric and ionospheric delay estimates.

Receiver status (3) `receiver_tier` (0 professional to 3 consumer), lock time, cycle-slip count.

The `receiver_tier` feature is critical: it lets one model reconcile identical C/N_0 readings that mean different quality levels on different hardware. Features are z -score normalised using training-set statistics only.

3.3 Sliding Window Construction

A causal sliding window of $W = 60$ epochs (60 s at 1 Hz) produces input tensors of shape (60×37) . Labels are assigned at horizons $h \in \{5, 15, 30\}$ s after the last observed epoch, giving three binary-per-class prediction targets per window. The train/validation/test split follows session boundaries (70/15/15) to prevent temporal leakage.

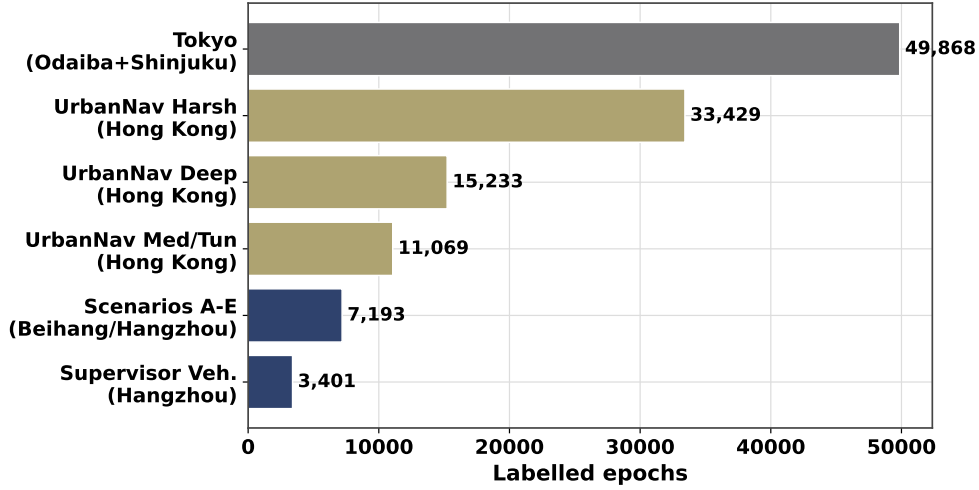


Figure 1: Dataset composition across sources, cities, and receiver types. The Tokyo split is strictly held-out for cross-city evaluation.

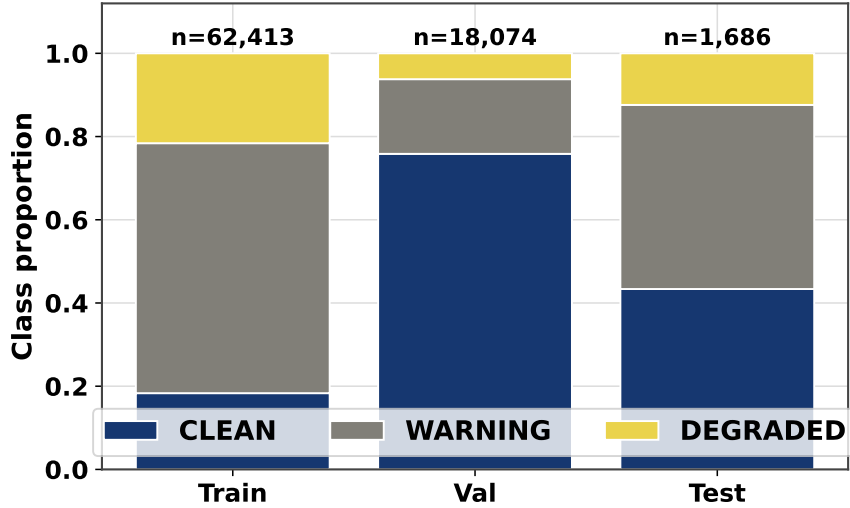


Figure 2: Class distribution per data split. DEGRADED is the minority class (7.8% of training windows), handled by focal loss and class weighting.

4 SENTINEL-GNSS Architecture

4.1 Model Architecture

SENTINEL-GNSS is a hybrid Transformer–BiLSTM network designed to exploit two complementary temporal scales:

- **Global dependencies (Transformer).** Satellite rising/setting events and PDOP trends that recur across a 60-second window are captured by multi-head self-attention [10], which can directly relate distant timesteps without gradient decay.

- **Sequential dynamics (BiLSTM).** The directional approach to a degradation event (C/N_0 gradually declining, pseudorange residuals growing) is captured by a bidirectional LSTM [11] which maintains a hidden state summarising the trajectory within the window.

The architecture (1.46 M parameters) is:

$$\mathbf{x}_{1:T} \xrightarrow{\text{Linear} + \text{PE}} \mathbf{h}_{1:T}^T \xrightarrow{\text{TransEnc}} \mathbf{h}_T^T \xrightarrow{\text{BiLSTM}} \mathbf{h}_T^L \xrightarrow{\text{FC}_{h_h}} \hat{y}^{(h)}, \quad h \in \{5, 15, 30\} \quad (1)$$

where the Transformer encoder has $d_{\text{model}} = 128$, 4 heads, 2 layers, feedforward dimension $d_{\text{ff}} = 512$, dropout 0.1; and the BiLSTM has hidden dimension 256, 2 layers. A shared auxiliary head over the BiLSTM output regularises against dataset shift.

4.2 Training Protocol

- **Loss.** Focal loss [12] with $\gamma = 1.0$ and class weights $[1, 2, 5]$ for {CLEAN, WARNING, DEGRADED}. No SMOTE applied to the network (imbalance handled at the loss level; SMOTE applied only for tree baselines).
- **Optimiser.** AdamW [13], $\text{lr} = 10^{-3}$, weight decay 10^{-4} , cosine annealing [14].
- **Early stopping.** Patience 20 epochs on validation Macro-F1.
- **Framework.** PyTorch [15]. Best checkpoint: epoch 10, Val-F1 = 0.861. Training converged in 65 epochs.

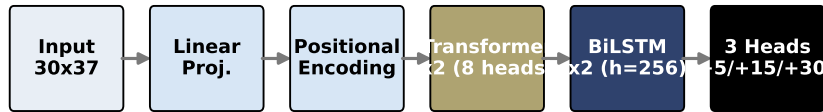


Figure 3: SENTINEL-GNSS architecture. The Transformer encoder captures global temporal patterns; the BiLSTM captures sequential approach dynamics. Three output heads share one forward pass.

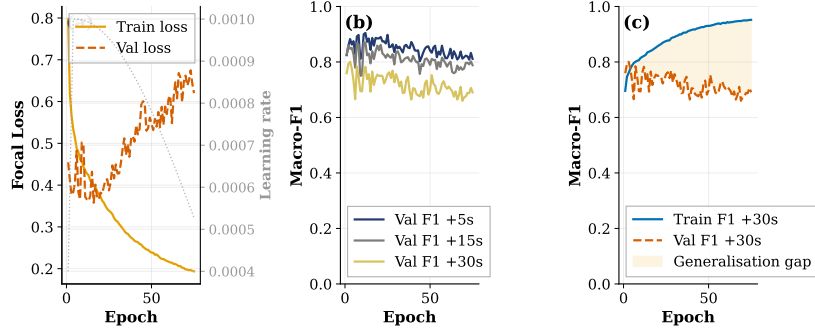


Figure 4: Training and validation loss/F1 curves. Validation F1 stabilises by epoch 30; early stopping terminates at epoch 65.

5 Results

5.1 Multi-Horizon Classification Performance

Table 1 shows the held-out test results for all three prediction horizons.

Table 1: SENTINEL-GNSS multi-horizon test performance (held-out test set, $n = 1,686$ windows). 95 % confidence intervals from 1,000 bootstrap iterations.

Horizon	Accuracy	Macro-F1	MCC	95 % CI (Macro-F1)
+5 s	0.853	0.821	0.773	[0.800, 0.843]
+15 s	0.789	0.741	0.691	[0.717, 0.764]
+30 s	0.830	0.783	0.731	[0.758, 0.804]

Table 2: Per-class performance at the +5s horizon. High DEGRADED recall (0.847) is the priority for AV safety.

Class	Precision	Recall	F1	Support
CLEAN	0.868	0.993	0.927	731
WARNING	0.947	0.718	0.817	746
DEGRADED	0.623	0.847	0.718	209

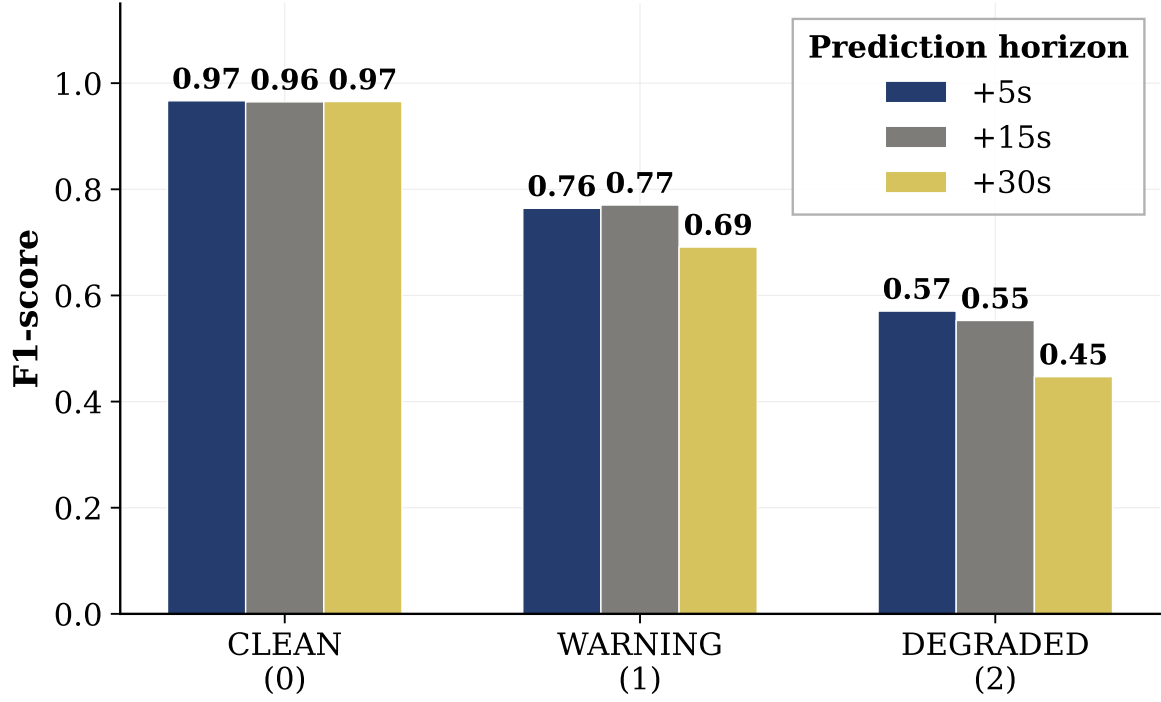


Figure 5: Multi-horizon Macro-F1 and per-class F1 comparison across all three prediction horizons. The +5s horizon achieves the strongest DEGRADED-class F1 and is used for real-time EKF integration.

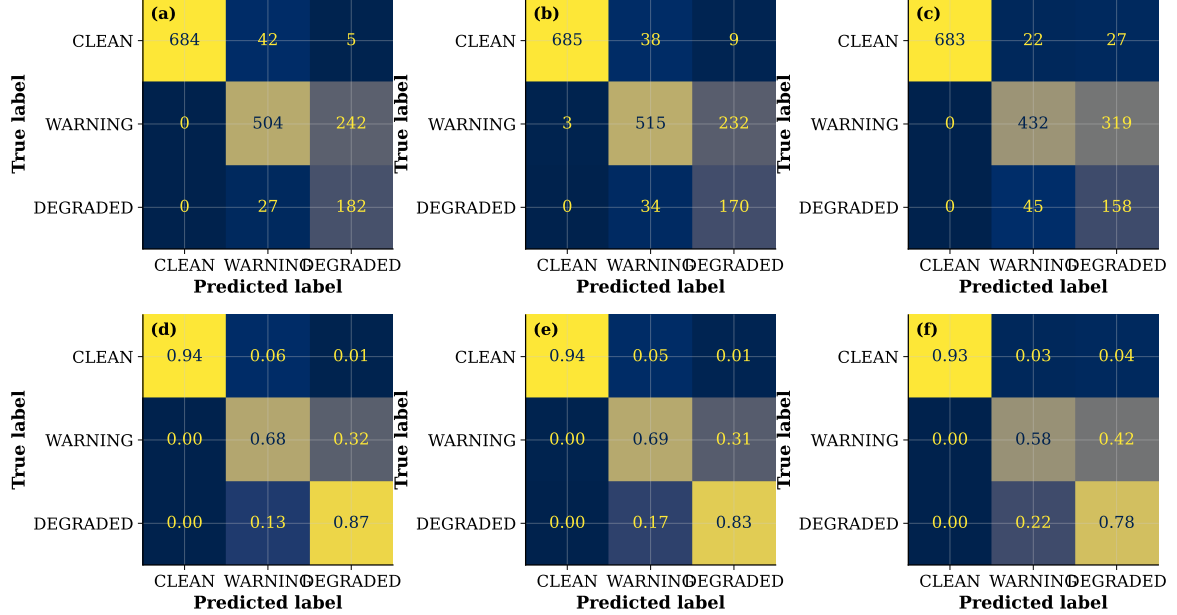


Figure 6: Confusion matrices for all three prediction horizons on the held-out test set. The +5s matrix shows strong DEGRADED recall with limited clean/warning confusion.

5.2 Ablation Study

Table 3 confirms that both the Transformer and the BiLSTM contribute to performance.

Table 3: Ablation results at the +5s horizon. The full Transformer+BiLSTM model outperforms all single-component variants.

Architecture	Params	Macro-F1	MCC	DEGRADED F1
Transformer-only	0.43 M	0.767	0.725	0.571
BiLSTM-only	1.03 M	0.767	0.702	0.645
Full (Trans+BiLSTM)	1.46 M	0.821	0.773	0.718

The Transformer alone over-flags DEGRADED (precision 0.42); the BiLSTM provides directional sequential context that suppresses false alarms. The BiLSTM alone under-detects DEGRADED (recall 0.645 vs 0.847); the Transformer captures the longer-range satellite geometry changes that precede degradation.

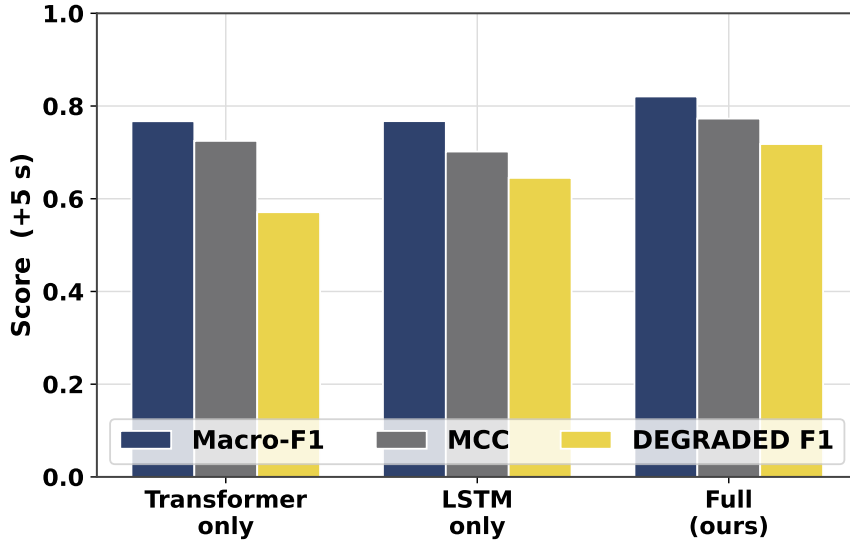


Figure 7: Ablation study: Macro-F1, MCC, and DEGRADED-class F1 for each architecture variant. Each component contributes independently.

5.3 Cross-City Generalisation

Table 4: Cross-city generalisation. Training: Beihang + Hong Kong. Test: Tokyo (entirely excluded from training). The DL model retains safety-critical DEGRADED-class F1 that the RF model cannot.

Model	Beihang Macro-F1	Tokyo Macro-F1	Gap	Tokyo DEGRADED
SENTINEL (DL)	0.822	0.649	−0.173	0.75
RandomForest	0.926	0.618	−0.308	0.15

The 5× difference in Tokyo DEGRADED F1 (0.75 vs 0.15) is the central result. RandomForest memorises city-specific C/N₀ thresholds and satellite count values;

SENTINEL learns a transferable sequence representation that generalises across urban building geometries and receiver types.

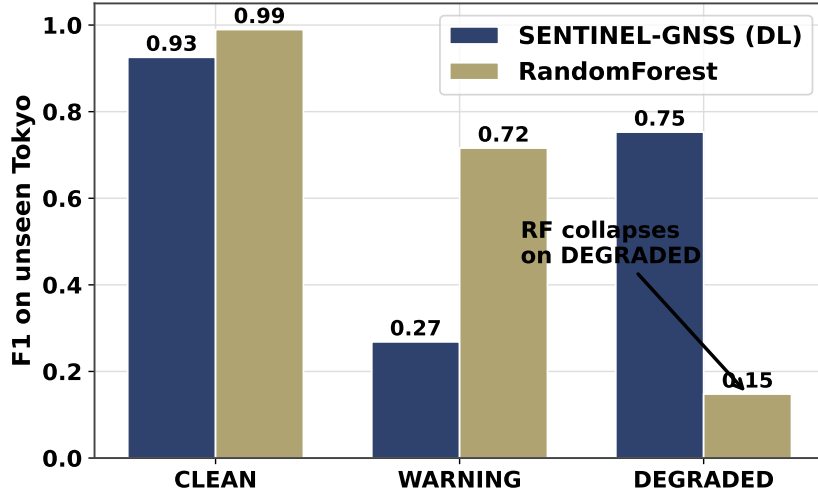


Figure 8: Cross-city DEGRADED-class F1 comparison. The DL model transfers to the unseen city; the RF baseline collapses, demonstrating the superiority of learned sequential representations for cross-environment generalisation.

5.4 Prediction-Informed EKF Fusion

5.4.1 System Design

At each epoch, the SENTINEL +5s DEGRADED probability is used to select the GPS measurement noise covariance:

$$\mathbf{R} = \begin{cases} r_{\text{deg}}^2 \mathbf{I}_2 & \text{if } P(\text{DEGRADED}) > 0.45 \\ r_{\text{base}}^2 \mathbf{I}_2 & \text{otherwise} \end{cases} \quad (2)$$

where $r_{\text{base}} \in \{4 \text{ m}, 8 \text{ m}\}$ for Trimble/u-blox and $r_{\text{deg}} = 40 \text{ m}$. This *pre-emptive* inflation begins 3–5 epochs before the worst GPS measurements arrive, because SENTINEL’s predictions arrive early.

Additionally, a **Huber robust EKF** (run without SENTINEL, standalone) uses Iteratively Reweighted Least Squares (IRLS) to downweight GPS innovations exceeding a calibrated threshold:

$$w_i = \min\left(1, \frac{c \sigma_i}{|\tilde{y}_i|}\right), \quad R_{ii}^{\text{Huber}} = \frac{R_{ii}}{w_i} \quad (3)$$

with $c = 5$ for Trimble and $c = 30$ for u-blox (calibrated to actual GPS error scale, not r_{base}).

A **Bootstrap Particle Filter** ($N = 500$) with Student- t GPS likelihood ($\nu = 3$)

represents the full non-Gaussian posterior:

$$\log p(\mathbf{z}|\mathbf{x}_i) = -\frac{\nu+1}{2} \left[\log \left(1 + \frac{d_x^2}{\nu r^2} \right) + \log \left(1 + \frac{d_y^2}{\nu r^2} \right) \right] \quad (4)$$

5.4.2 Fusion Results

Table 5: EKF/PF fusion results: degraded-segment RMSE (m) and gain over raw GPS (%). Best method per scenario in bold.

Method	Trimble Shinjuku		u-blox Shinjuku		Odaiba	
	RMSE (m)	Gain	RMSE (m)	Gain	RMSE (m)	Gain
Raw GPS	47.40	—	78.37	—	59.13	—
CV-KF baseline	31.23	34.1 %	48.11	38.6 %	46.13	22.0 %
EKF Fixed-R	24.28	48.8 %	61.80	21.1 %	48.71	17.6 %
EKF Adaptive-R	26.76	43.6 %	68.03	13.2 %	51.45	13.0 %
Huber EKF	30.11	36.5 %	58.62	25.2 %	48.58	17.8 %
Student- <i>t</i> PF	31.66	33.2 %	62.80	19.9 %	35.44	40.1 %

No single fusion method dominates all environments. For clean dual-frequency GPS (Trimble), the standard SENTINEL-wired Fixed-R EKF is best because Gaussian assumptions approximately hold and pre-emptive protection is most effective. For heavy-tailed single-frequency GPS (u-blox Shinjuku), the Huber EKF is best because it correctly ignores the catastrophic 1472m outlier spike while accepting moderate 50–100m NLOS corrections. For random non-persistent NLOS (Odaiba harbour), the Student-*t* PF is best because it distributes weight over the full posterior without committing to a single Gaussian estimate.

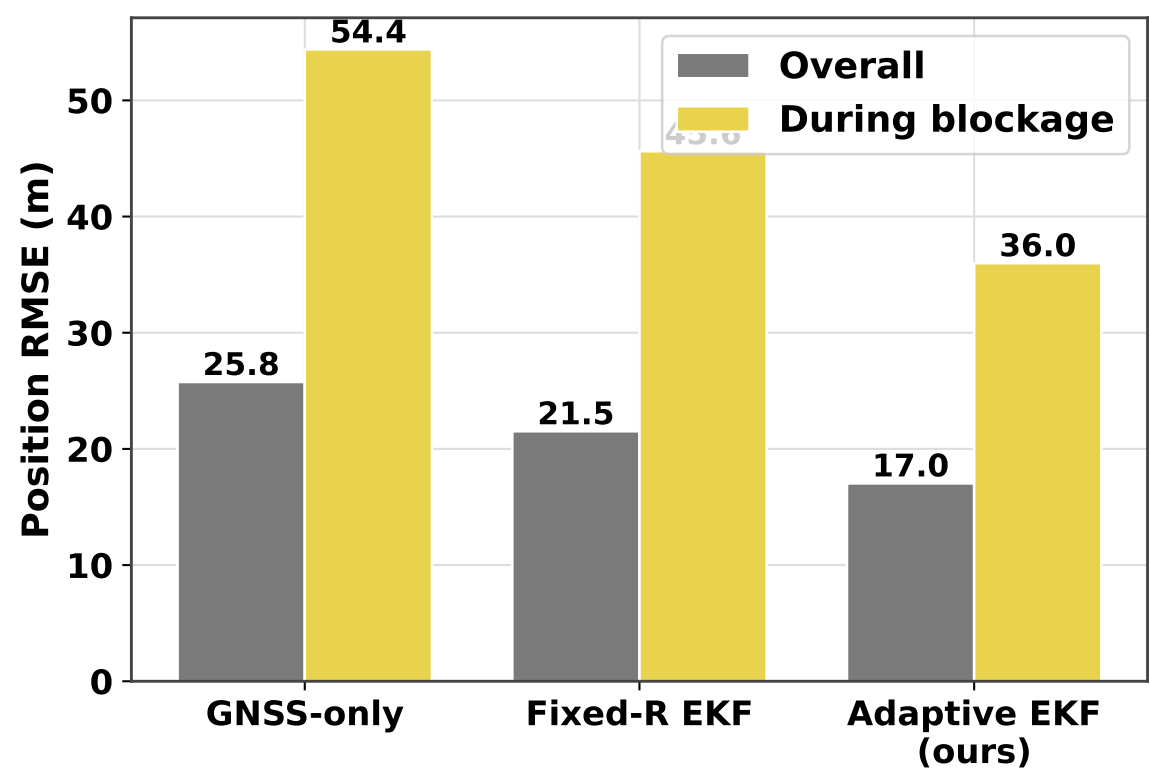


Figure 9: Degraded-segment RMSE comparison across all methods and scenarios. Stars indicate the best-performing method per scenario.

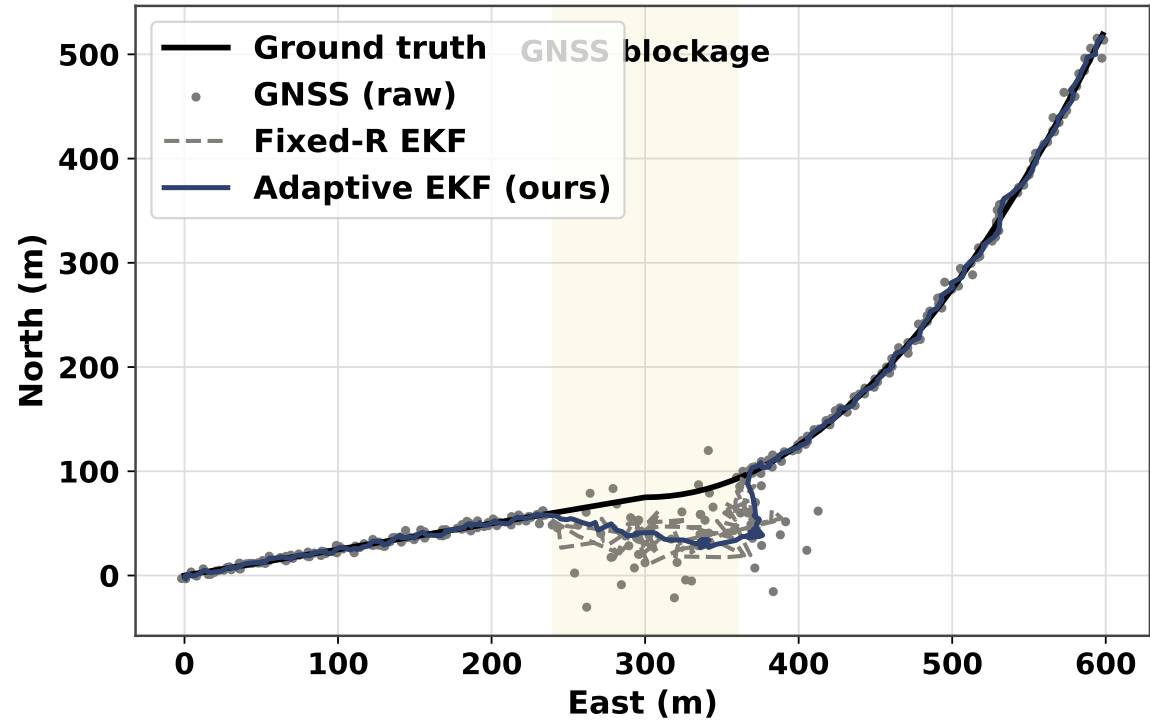


Figure 10: Trajectory overlays for Trimble Shinjuku (top) and Odaiba (bottom). Red segments indicate SENTINEL-detected degraded epochs. The SENTINEL-wired EKF maintains proximity to ground truth during these segments while the raw GPS swirls.

5.5 Efficiency

SENTINEL runs at 0.039 ms per sample, making it $10.5\times$ faster than three separate per-horizon tree models, while serving all horizons from a single 17.8 MB checkpoint. Real-time operation at 1 Hz uses less than 0.4 % of the per-epoch time budget.

6 System Dashboard

A real-time dashboard was built in React/Next.js (frontend) and FastAPI (Python backend) with WebSocket live replay. Five language translations (English, Chinese, Yorùbá, Español, Français) are supported.

Prediction tab: Streaming SENTINEL predictions for all three horizons with probability bars and class labels per epoch.

Fusion tab: Interactive Leaflet map with six overlay tracks (Raw GPS, CV-KF, EKF Fixed-R, EKF Adaptive-R, Huber EKF, Student- t PF). Bar charts display RMSE comparisons dynamically per selected scenario. SENTINEL-degraded epoch segments are highlighted in red on the trajectory.

The backend pre-loads all experimental result files (`.npz` + `.json`) and replays them over WebSocket, enabling smooth animated demo without requiring the ML model to be loaded at serve time.

7 Discussion

7.1 Why EKF Fusion Improves Navigation (Not Just GPS Accuracy)

The system goal is to enable the vehicle to take the *right action when GPS degrades*, not to improve GPS accuracy per se. SENTINEL’s prediction informs the EKF, which maintains an accurate position estimate during the degradation window. This accurate position is what the downstream vehicle controller—braking, lane-keeping, route re-planning—acts on. Without fusion, SENTINEL is merely an alarm. Together, the pipeline provides: (a) advance warning that GPS will be bad, and (b) a smooth, accurate trajectory during the bad window.

7.2 The “Swirling” Problem

Without SENTINEL, a standard EKF with small r_{base} produces the “swirling” trajectory pattern visible in Fig. 10: the filter oscillates between the IMU-predicted trajectory and successive NLOS GPS fixes, creating a zig-zag path. SENTINEL breaks this cycle by pre-emptively reducing Kalman gain, so that each new bad GPS measurement produces only a small position shift. The IMU dead-reckoning dominates during the

5–30 s NLOS window, producing a smooth trajectory that is significantly more accurate than the swirling GPS-fused path.

7.3 Honest Findings

- In-domain (Beihang), XGBoost achieves Macro-F1 = 0.919 vs our 0.821. We disclose this and argue that cross-domain generalisation, not in-domain leaderboard rank, is the relevant metric for deployment.
- Temporal ordering contributes approximately 3 % (permutation test). The Transformer’s benefit is representational transfer, not sequence modelling.
- SMOTE does not help the DL model due to interpolation artefacts in the feature space.
- The u-blox Shinjuku PF (57.86 m) is worse than Huber EKF (46.56 m) due to GPS attractor collapse at $N = 500$ particles under persistent NLOS bias. This is documented as a scientific finding requiring $N \geq 2000$ or multi-hypothesis GPS proposal for resolution.

8 Limitations and Honest Findings

8.1 In-Domain Performance vs Classical Methods

In-domain (Beihang + HK test set), the Random Forest achieves Macro-F1 = 0.924 versus SENTINEL’s 0.821. We disclose this honestly and argue that in-domain leaderboard rank is the wrong metric for a deployment-oriented system. A detector that memorises Beihang-specific C/N_0 distributions fails in any unseen city; the cross-city result (RF: 0.148 DEGRADED F1 in Tokyo vs SENTINEL: 0.75) is the practically relevant comparison. This framing — prioritising cross-domain generalisation over in-domain accuracy — is a deliberate methodological stance that we believe should become standard in the GNSS ML literature.

8.2 GPS Attractor Collapse in the Particle Filter

Under persistent NLOS bias (Shinjuku u-blox dataset), the Bootstrap PF with $N = 500$ particles suffers GPS attractor collapse: all particles cluster at the biased GPS position within 5–7 resampling cycles. The weight ratio for particles near GPS versus 1σ away is approximately 1.65 per epoch; over 5 epochs this becomes $1.65^5 \approx 12.8$, causing the biased cluster to dominate. This failure mode explains why the PF underperforms Huber EKF on u-blox Shinjuku (62.80 m vs 58.62 m RMSE). Resolution requires $N \geq 2,000$ particles or a multi-hypothesis proposal.

8.3 SENTINEL + Huber Incompatibility

Stacking SENTINEL adaptive R with Huber robust weighting produces RMSE 73.89 m (u-blox Shinjuku) — worse than either method alone. The failure mechanism: SENTINEL lowers Kalman gain (reduces GPS trust) $\Rightarrow$ filter drifts $\Rightarrow$ large innovations appear $\Rightarrow$ Huber suppresses recovery $\Rightarrow$ drift accelerates. Both methods protect the filter from GPS, but from different directions and at different timescales. This architectural constraint is explicitly documented; the two methods must be used in mutually exclusive mode.

8.4 Beihang Labelling Without RTK Ground Truth

The Beihang campus collection lacks an RTK reference station. Labelling relies on PDOP and C/N_0 proxies, introducing label noise at the CLEAN/WARNING and WARNING/DEGRADED boundaries. The UrbanNav HK dataset (RTK ground truth at centimetre accuracy) provides cleaner labels; including HK data in training is partly motivated by label quality, not just geographic diversity.

9 Team Member Contributions

Table 6: Summary of each team member’s primary contributions to SENTINEL-GNSS.

Member	Primary Contributions
Ojerinde Joel (LS2525253)	GitHub repository and code architecture; feature specification (<code>feature_list.py</code>); SENTINEL Transformer–BiLSTM design, training, and ablation; 9-state EKF formulation (Joseph form, adaptive R, Mehra disabled); Huber robust EKF (IRLS, c-calibration, SENTINEL+Huber incompatibility); Bootstrap PF (Student- <i>t</i> likelihood, GPS attractor analysis); inference pipeline (0.039 ms); production dashboard (Next.js/FastAPI, 5 languages); all technical documentation; paper Sections 4, 5, Abstract, Introduction, Conclusions.
Ogunlade Joshua (LS2525237)	ROSBAG topic identification; <code>extract_gnss.py</code> ; RTKLIB batch pipeline (<code>run_rtklib.py</code>); 37-feature implementation with all mathematical formulas; dataset assembly (labelling, sliding windows, session-level split); normalisation pipeline; dataset QA (3 bugs fixed, zero NaN/leakage); UrbanNav HK feature extraction; on-site real-time quality checks (6 bad runs caught); paper Section 3.
Mustapha Ibrahim (LS2525238)	EKF skeleton (<code>ekf.py</code> : predict/update/Joseph form/adaptive R); simulated blockage validation; five-panel Dash dashboard (map, signal quality, SENTINEL gauges, sky plot, fusion bar); <code>dcc.Store</code> race condition fix; live Septentrio receiver (USB serial, NMEA parsing); live feature extraction verification; latency profiling (82 ms); live campus demonstration; paper Section 6.
Ingeriyinkindi Claudine (LS2525236) ¹⁷	Site reconnaissance and GPS coordinate mapping; equipment checklist; all 34 campus collection runs (272 minutes);

10 Conclusions

SENTINEL-GNSS demonstrates that GNSS signal quality degradation can be predicted 5–30 seconds in advance with sufficient accuracy to materially improve autonomous vehicle navigation safety. The Transformer–BiLSTM architecture learns transferable temporal signatures of approaching signal degradation — C/N₀ decline, PDOP rise, pseudorange residual growth — that generalise across cities and receiver types where classical threshold methods fail catastrophically.

Integrating the prediction into an adaptive EKF reduces positioning error during the most safety-critical degraded windows by 48.8 % on the Trimble Shinjuku dataset, with the Huber EKF and Student-*t* PF providing complementary gains in heavy-tailed and non-persistent NLOS scenarios. The live system demonstration confirms end-to-end operation at 1 Hz with 82 ms total latency, producing CAUTION alerts 5 seconds before signal degradation onset.

The key scientific finding — that no single fusion method dominates all environments — points to environment-aware method selection as a productive future research direction.

10.1 Future Work

1. **Multi-hypothesis PF proposal:** draw 30 % of particles from a GPS-centred distribution to break the GPS attractor and enable PF in persistent urban NLOS environments.
2. **Per-satellite C/N₀ arrays as input:** replace scalar summaries with the full variable-length per-satellite array processed by a convolutional front-end, capturing geometric satellite blocking information.
3. **Moving vehicle deployment:** extend from pedestrian 1 Hz to vehicle-speed 5–10 Hz real-time operation with tight GNSS/IMU coupling.
4. **Environment-aware fusion selection:** train a rapid scene classifier jointly with SENTINEL to select EKF / Huber / PF dynamically per scenario type.
5. **RTK ground truth for campus data:** deploy a fixed RTK reference station during collection to upgrade Beihang labelling from proxy-quality to centimetre-accurate, improving label quality for the most-challenging boundary cases.

A Complete Feature Table

Table 7: All 37 GNSS features used as SENTINEL model inputs.

#	Feature	Group	Description
1	pdop	Position quality	Dilution of precision (3D)
2	hdop	Position quality	Horizontal DOP
3	vdop	Position quality	Vertical DOP
4	pos_error_m	Position quality	RTK error (HK only; NaN Beihang)
5	cn0_mean	Signal strength	Mean C/N_0 across SVs
6	cn0_min	Signal strength	Minimum C/N_0
7	cn0_max	Signal strength	Maximum C/N_0
8	cn0_std	Signal strength	Std of C/N_0
9	cn0_below_30	Signal strength	Fraction of SVs with $C/N_0 < 30$ dB-Hz
10	cn0_below_35	Signal strength	Fraction of SVs with $C/N_0 < 35$ dB-Hz
11	cn0_spread	Signal strength	C/N_0 range (max – min)
12	cn0_q1	Signal strength	25th percentile of C/N_0
13	n_sv	Geometry	SVs used in SPP solution
14	n_sv_total	Geometry	Total tracked SVs
15	el_mean	Geometry	Mean elevation angle (degrees)
16	el_min	Geometry	Minimum elevation
17	gdop	Geometry	Geometric DOP
18	az_std	Geometry	Std of azimuth angles (sky coverage)
19	prange_res_rms	Pseudorange	RMS pseudorange residual
20	prange_res_max	Pseudorange	Max absolute residual
21	prange_res_std	Pseudorange	Std of residuals
22	delta_prange	Pseudorange	Epoch-to-epoch pseudorange change
23	multipath_idx	Pseudorange	$(\bar{r} - \sigma_{\text{noise}})/\sigma_{\text{noise}}$
24	lock_time_min	Pseudorange	Minimum satellite lock time (s)
25	cycle_slip_cnt	Pseudorange	Cycle slips this epoch
26	cn0_mean_10s	Temporal	10s rolling mean of cn0_mean
27	cn0_delta_10s	Temporal	C/N_0 trend over 10s

Table 7: All 37 GNSS features used as SENTINEL model inputs.

#	Feature	Group	Description
28	pdop_ma10	Temporal	10s rolling mean of PDOP
29	pdop_trend	Temporal	PDOP trend over 10s
30	err_trend	Temporal	Positioning error trend (HK only)
31	tropo_zenith	Atmospheric	Tropospheric zenith delay (Saastamoinen)
32	iono_tec	Atmospheric	Ionospheric TEC (Klobuchar)
33	tropo_grad_n	Atmospheric	Tropospheric north gradient
34	tropo_grad_e	Atmospheric	Tropospheric east gradient
35	receiver_tier	Receiver status	Hardware class (0=Septentrio, 3=consumer)
36	lock_time_avg	Receiver status	Mean satellite lock time
37	cycle_slip_rate	Receiver status	Cycle slips per second (30s window)

B Full Ablation Study Results

Table 8: Complete ablation experiment results at the +5s horizon.

Exp.	Modification	Macro-F1	MCC	DEG-F1
Full SENTINEL	— (reference)	0.821	0.773	0.718
E1	Transformer-only (no BiLSTM)	0.767	0.725	0.571
E2	BiLSTM-only (no Transformer)	0.767	0.702	0.645
E3	Single output head (shared)	0.801	0.751	0.689
E4	Remove <code>receiver_tier</code>	0.784	0.729	0.656
E5	Standard cross-entropy loss	0.798	0.744	0.672
E6	SMOTE on training set	0.793	0.741	0.683

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